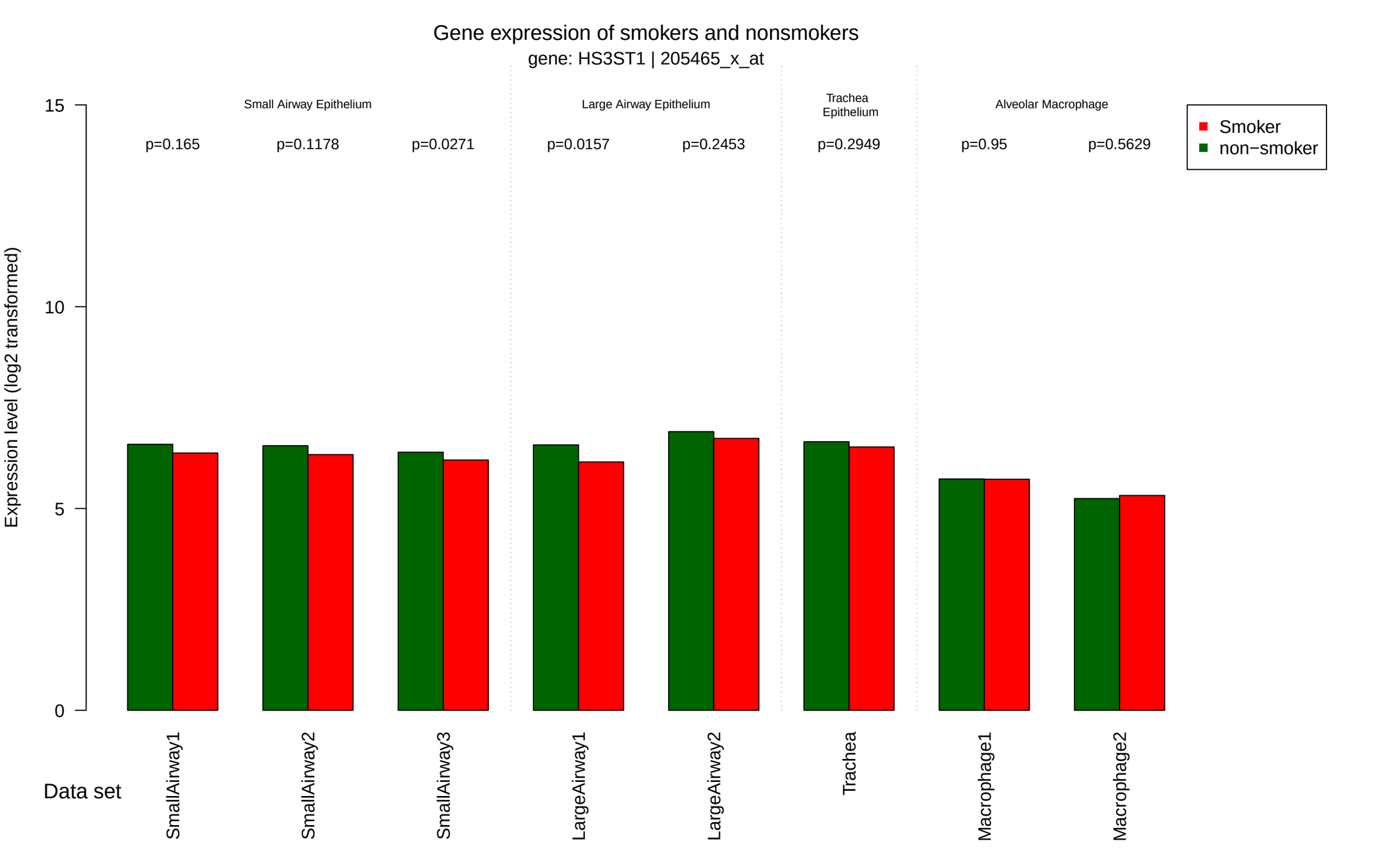
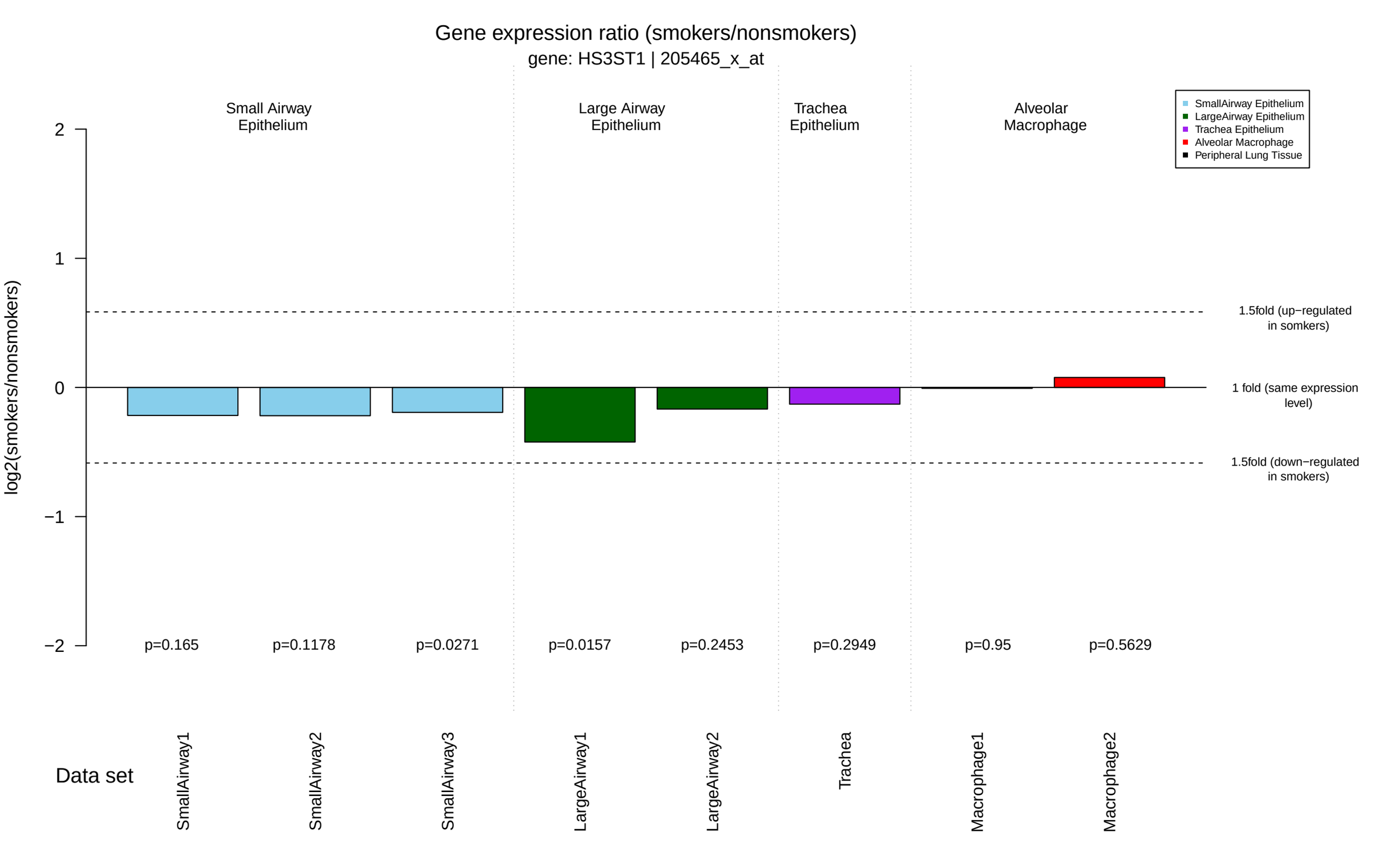
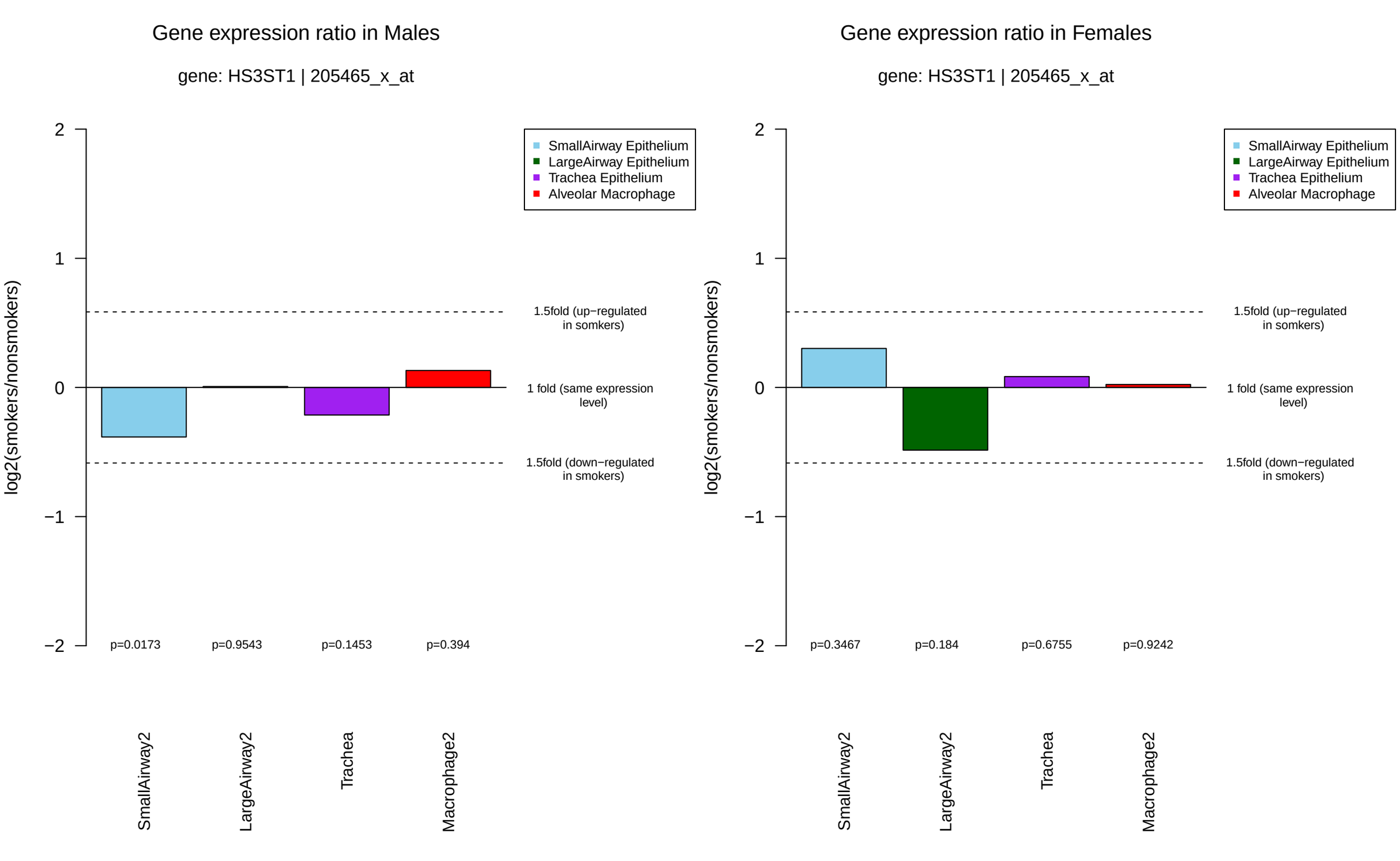
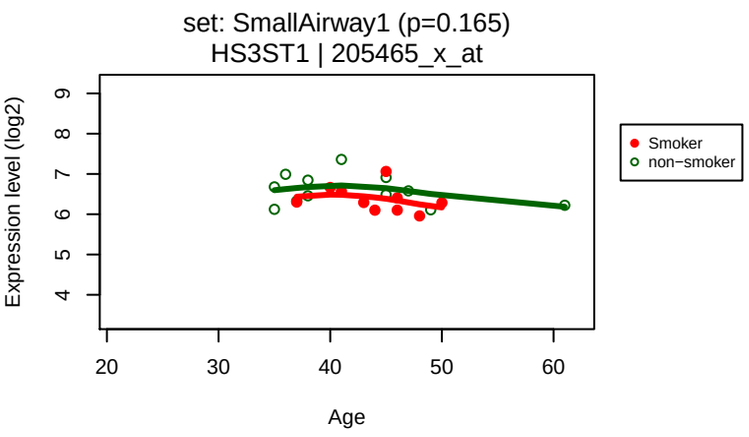
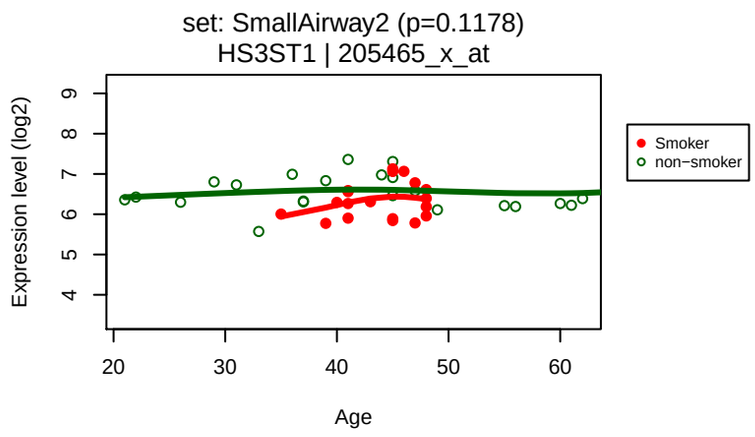
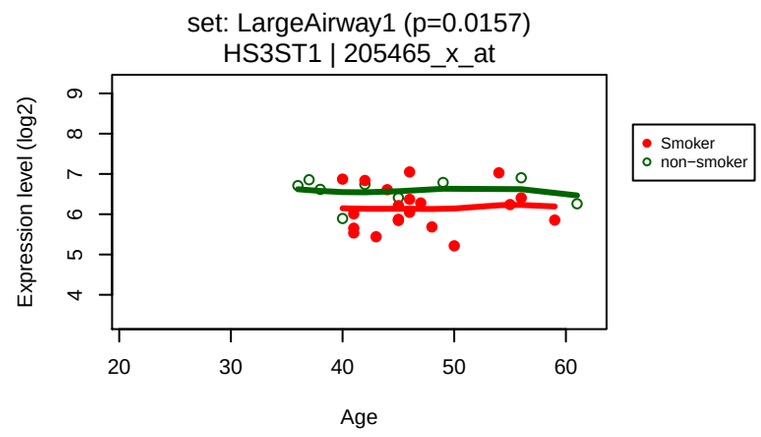

Gene expression of smokers and nonsmokers
gene: HS3ST1 | 205465_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.165
p=0.1178
p=0.0271
p=0.0157
p=0.2453
p=0.2949
p=0.95
p=0.5629
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

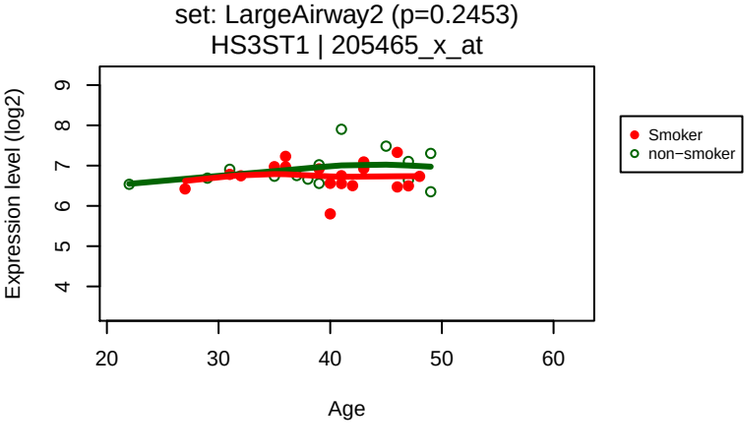

Gene expression ratio (smokers/nonsmokers)
gene: HS3ST1 | 205465_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.165
p=0.1178
p=0.0271
p=0.0157
p=0.2453
p=0.2949
p=0.95
p=0.5629
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

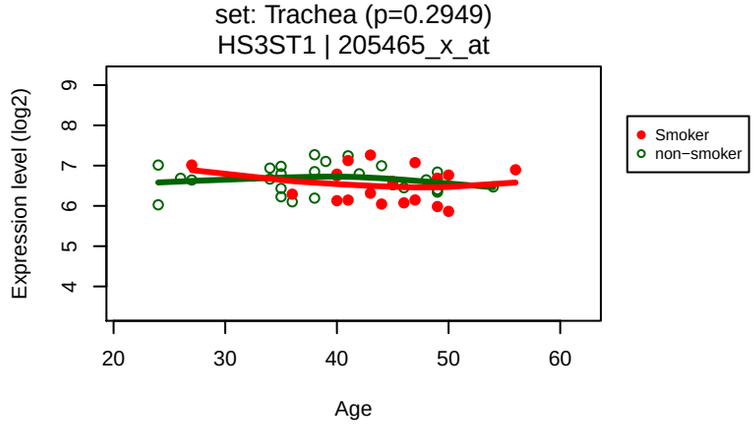

Gene expression ratio in Males
gene: HS3ST1 | 205465_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0173
p=0.9543
p=0.1453
p=0.394
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HS3ST1 | 205465_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3467
p=0.184
p=0.6755
p=0.9242
SmallAirway2
LargeAirway2
Trachea
Macrophage2

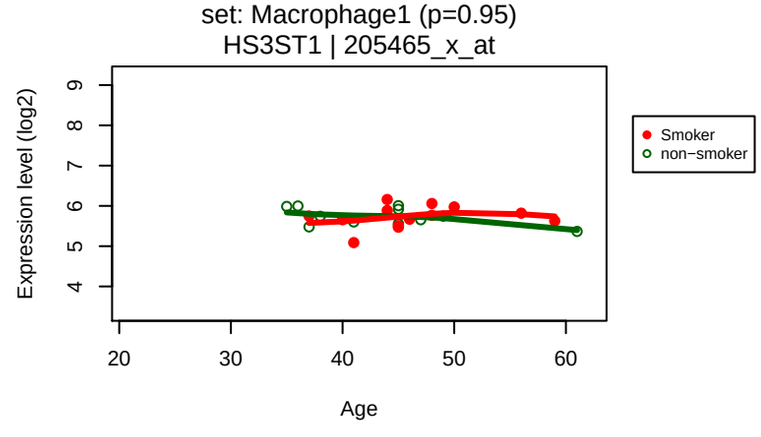

set: SmallAirway1 (p=0.165)
HS3ST1 | 205465_x_at
Expression level (log2)
Age
Smoker
non−smoker

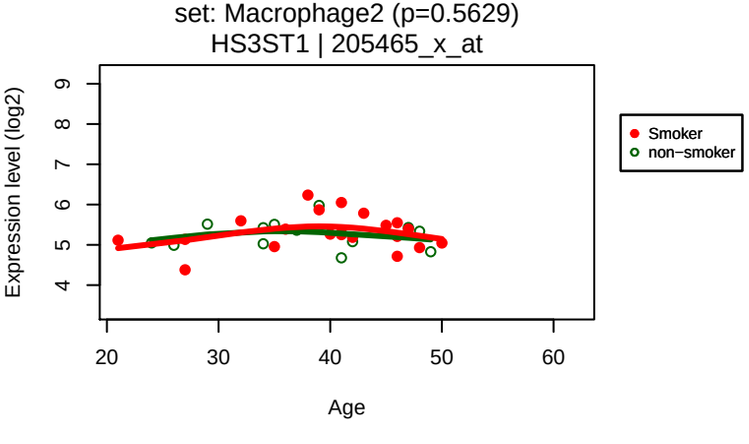

set: SmallAirway2 (p=0.1178)
HS3ST1 | 205465_x_at
Expression level (log2)
Age
Smoker
non−smoker

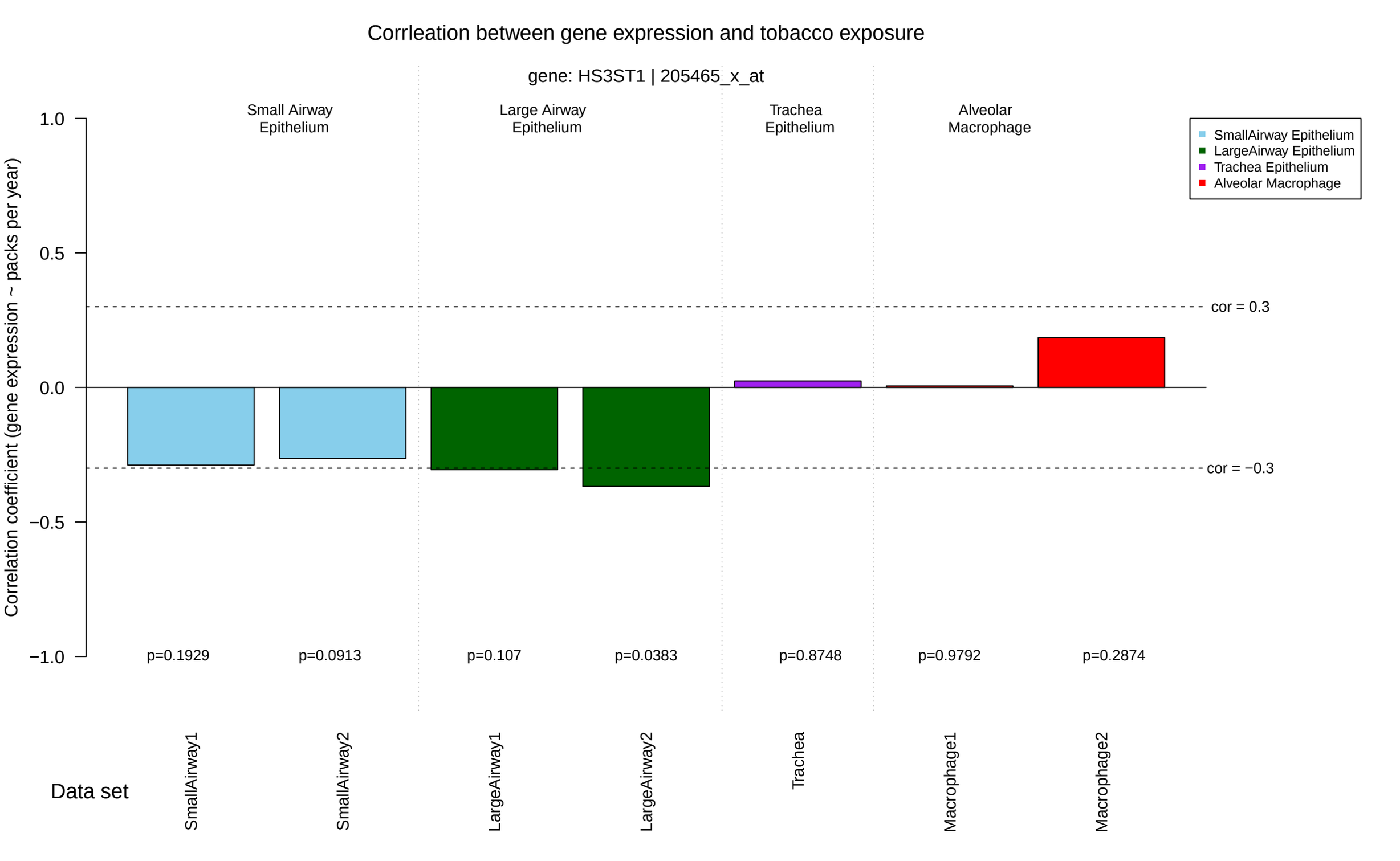

set: LargeAirway1 (p=0.0157)
HS3ST1 | 205465_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2453)
HS3ST1 | 205465_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2949)
HS3ST1 | 205465_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.95)
HS3ST1 | 205465_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5629)
HS3ST1 | 205465_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HS3ST1 | 205465_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1929
p=0.0913
p=0.107
p=0.0383
p=0.8748
p=0.9792
p=0.2874
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2